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REVIEW

The effects of inulin on gut microbial composition: a systematic review of evidence from human studies

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Abstract

Background Inulin, consisting of repetitive fructosyl units linked by β(2,1) bonds, is a readily fermentable fiber by intestinal bacteria that generates large quantities of short-chain fatty acids (SCFA). In individuals with constipation, it was reported that inulin ingestion was associated with a significant increase in stool frequency, suggesting a potential impact of inulin on human gut microbiota composition. Progress in high-throughput technologies allow assessment of human-associated microbiomes in terms of diversity and taxonomic or functional composition, and can identify changes in response to a specific supplementation. Hence, to understand the effects of inulin on the human gut microbiome is pivotal to gain insight into their mechanisms of action.

Methods Here, we conducted a systematic review of human studies in adult individuals showing the effects of inulin on the gut microbiome. We searched in MEDLINE, EMBASE, Web of Science, and Scopus databases for articles in English published in peer-reviewed journals and indexed up until March 2019. We used multiple search terms capturing gut microbiome, gut microflora, intestinal microbiota, intestinal flora, gut microbiota, gut flora, microbial gut community, gut microbial composition, and inulin.

Results Overall, nine original articles reported the effects of inulin on microbiome composition in adult humans, most of them being randomized, double-blind, placebo-controlled trials (n = 7). Studies varied significantly in design (3 studies associated inulin and oligofructose), supplementation protocols (from 5 to 20 gr per day of inulin consumed) and in microbiome assessment methods (16S sequencing, n = 7). The most consistent change was an increase in *Bifidobacterium*. Other concordant results included an increase in relative abundance of *Anaerostipes*, *Faecalibacterium*, and *Lactobacillus*, and a decrease in relative abundance of *Bacteroides* after inulin supplementation.

Conclusions Our systematic review assessed the evidence for the effects of inulin supplementation on the human gut microbiome. However, these in vivo studies did not confirm in vitro experiments as the taxonomic alterations were not associated with increase in short-chain fatty acids levels.

Keywords Inulin · Prebiotic · Gut microbiome · Diversity · Short-chain fatty acids

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Background

Prebiotic are defined as non-digestible food ingredient that beneficially affects the host by selectively stimulating the growth and/or activity of one or a limited number of bacteria in the colon, and thus improves health [1]. They are referenced since 1995. Inulin, consisting of repetitive fructosyl units linked by β(2,1) bonds [2], is a readily fermentable fiber by intestinal bacteria, generating large quantities of short-chain fatty acids (SCFA) [3]. To date, a significant increase in stool frequency following inulin ingestion was reported in healthy individuals with constipation [4]. In 2015, European Food Safety Authority stated that inulin contributes to maintenance of normal defecation by increasing stool frequency, based on a daily consumption of 12 g of native chicory inulin [5]. However, the American College of Gastroenterology suggested against the use of prebiotics for overall symptom improvement in patients with inflammatory bowel syndrome (IBS) [6]. Moreover, Singh et al showed that replacing cellulose by inulin exacerbate αIL10R-induced colitis in mice, by enrichment of butyrate-producing bacteria in the gut. Thus, prebiotic can also lead to gut dysbiosis and worsen inflammatory bowel diseases (IBD) [7]. A recent systematic review also reported that serious adverse events in clinical trials assessing prebiotics, or other interventions to manipulate the gut microbiome, are often not reported or inadequately measured [8]. Furthermore, progress in high-throughput technologies allow assessment of human-associated microbiomes in terms of composition and function, and can identify specific changes in response to a specific supplementation [9]. Thus, it is pivotal to understand the effects of inulin on human gut microbiome to gain insight into their mechanisms of action. Hence, we sought to systematically review current evidence of inulin effect on the human gut microbiome.

Methods

Search strategy

We followed the recommended Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines to select the articles. The methods who applied here are consistent with those of previously published systematic reviews in the field of human microbiome [10, 11]. We searched in MEDLINE, Web of Science, and Scopus databases for articles published in peer-reviewed journals, written in English, and indexed up until March 2019. We used several search terms including gut microbiome, gut microflora, intestinal microbiota, intestinal flora, gut microbiota, gut flora, microbial gut community, gut microbial composition and inulin. The search strategy is detailed in Additional file 1.

Study population and interventions included in the review

We included observational, prospective or retrospective, cross-sectional, case-control, cohort studies or intervention studies involving adult humans (age 18 years or older), that reported effects of inulin on the gut microbiome. Any comparator was permitted. We excluded studies that included preclinical models and in vitro studies that used fecal material from healthy volunteers and inulin growth media.

Outcomes studied

We assessed differences in the human gut microbiome in terms of alpha and beta diversity, and in terms of taxonomy, after exposure to inulin, compared with baseline value or to control arm.

Study selection and data extraction

All titles and abstracts found by the search strategy as reported in Additional file 1 were screened for relevance by EM. Titles and abstracts were then retrieved in full and evaluated for inclusion eligibility in the systematic review by E.M. and Q.L.B., independently. Eligibility results were compared and discussed between Q.L.B, G.C., F.J, M.D, and E.M. to finally retain 9 studies for report in the systematic review, as detailed in Figure 1. Data were extracted into pre-specified structured tables. Missing data points were marked “unknown” or “not reported”. We did not request additional data from the authors of the published reports, and did not analyze data not included in the articles selected in the systematic review.

Summary measures

In the first pre-specified table, we detailed the compound studied and the dosage, the comparator used, the duration of exposure of the treatment, the number of participants in the study, and the microbiome assessment method applied in the study [10]. We also performed a qualitative synthesis of findings from different studies published and included in the systematic review. In the second pre-specified table, we reported the total gut bacterial diversity and the taxonomic changes observed in the published reports. In detail, alpha diversity, defined as diversity within each individual sample [12], was reported using one or several metrics that can be used: Shannon index, Chao1 index, Simpson index, operational taxonomic unit (OTU) or species count or phylogenetic diversity. Importantly, an OTU is a population of phylogenetically related organisms, grouped on the basis of 16S RNA sequence similarity. Usually, a 97% similarity level is used to group similar sequences into an OTU [12]. Beta diversity that represents the intervariability between samples and describes

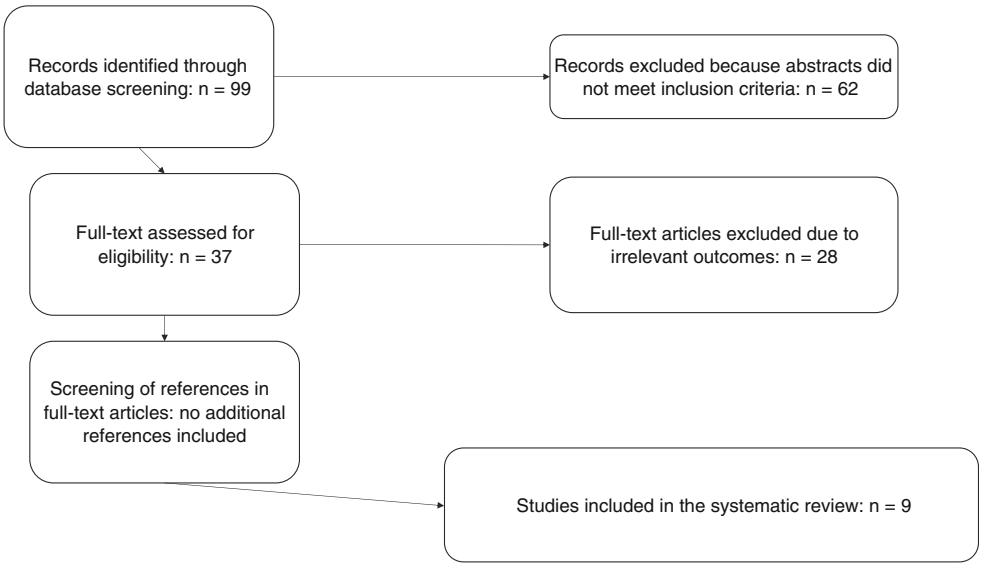

Fig. 1 Study for diagram.

how samples cluster together was reported using Bray Curtis or UniFrac distances matrix [13]. Taxonomy is usually reported with relative abundance that is the percent composition of a specific taxon, relative to the total number of taxa the gut microbiome. Here, we formatted results to universal taxonomy from phylum level to lowest available taxonomic level (i.e., genus or species) using the NCBI Taxonomy browser [10, 14]. We summarized relative abundance at various taxonomic levels as "increased," "decreased," or "unchanged as previously reported [10].

Results

Study selection

Overall, ninety nine relevant abstracts were identified through MEDLINE, Web of Science, and Scopus with our search strategy (Fig. 1). We did not find any review articles in this selection.

Study characteristics

Of the nine included studies, all were randomized and controlled trials [15–23]. Overall, published reported varied significantly in terms of design that may explain variability in the response to inulin supplementation (Table 1). Four were crossover trials, with washout periods varying from 7 days to 3 weeks after the first intervention [15, 16, 18, 22]. The studies included healthy subjects ($n = 5$) [15–18, 22], overweighed or obese subjects ($n = 2$) [19, 20], and patients with IBD, including active (mild to moderately) or inactive disease ($n = 2$) [21, 23]. In one study, participants were categorized with low (< 18 g/day for females and < 22 g/day for males) or high (≥ 25 g/day for females and ≥ 30 g/day for males) dietary fiber intakes [18].

Daily supplementation with inulin was highly variable between the studies, from 5 grams per day [16] to 20 g/day [17, 23], and sources of inulin also varied (i.e., inulin isolated from chicory root, isolated from agave, extracted from globe artichoke). Importantly, six studies consisted in a manufactured and combined prebiotic supplementation (manufactured Orafti® Synergy1). Seven studies used 16S sequencing to report gut microbiota changes in diversity or taxonomy, and 4 applied the QIIME software to analysis sequences with the Greengenes core reference database. Overall, due to the substantial differences in design, supplementation dosages and the reporting of outcomes, pooling of results was not feasible, we were only able to analyze common trends in term of changes in diversity and taxonomy.

Results of individual studies

Five studies reported the effect of inulin on alpha diversity, using various metrics, including Chao1 index, Shannon index, inverse Simpson index or the number of observed species [15–19] (Table 2). In detail, four studies did not find significant changes in alpha diversity of the gut microbiome following inulin supplementation, compared to baseline or control arm [15–18]. One study reported a decrease in alpha diversity using the observed species metric, but no significant change using the Chao1 index [19]. Six studies assessed the effect of inulin on beta diversity (Table 2). In detail, two studies did not find any significant changes in the overall gut microbiome architecture [16, 21], whereas three studied reported significant change following inulin supplementation [15, 17, 19]. Importantly, one study highlighted a large inter-individual variation in the architecture of the gut microbiome in response to inulin supplementation, with participants having their pre and

Table 1 Description of included studies

	Title	Design	Compound studied and dose	Comparator
Vandeputte et al. 2017	Prebiotic inulin-type fructans induce specific changes in the human gut microbiota	Randomized, double-blind, placebo-controlled cross-over trial Preceded by a 2-week run-in phase during which the subjects were administered the placebo control. Washout phase after the first intervention of 2 weeks	Inulin 12 g/day chicory-derived inulin (Orafti® inulin)	Maltodextrin
Holscher et al. 2015	Agave inulin supplementation affects the fecal microbiota of healthy adults participating in a randomized, double-blind, placebo-controlled, crossover trial	Randomized, double-blind, placebo-controlled 7-day washouts between periods Crossover trial	Chocolate chews (Bruce's Candy Kitchen) providing 5.0 or 7.5 g agave inulin	Chocolate chews (Bruce's Candy Kitchen)
Baxter et al. 2019	Dynamics of human gut microbiota and short-chain fatty acids in response to dietary interventions with three fermentable fibers	Randomized study Four separate academic semesters During the first semester, participants blinded to the supplement they were taking; in subsequent studies, they were informed of its identity. Researchers analyzing samples were always blinded to the supplement associated with the samples.	Inulin isolated from chicory root (Swanson Health Products, Fargo, ND) 20 g/day	Maize starch
Healey et al. 2018	Habitual dietary fiber intake influences gut microbiota response to an inulin-type fructan prebiotic: a randomized, double-blind, placebo-controlled, cross-over, human intervention study	Randomized, double-blind, placebo-controlled, cross-over study 3-week washout period	Orafti® Synergy 1; 16 g/day (Orafti® Synergy1—50:50 inulin to fructooligosaccharide mix; Beneo GmbH) chicory-derived inulin	Glucidex® 29 Premium; 16 g/day
Reimer et al. 2017	Inulin-type fructans and whey protein both modulate appetite but only fructans alter gut microbiota in adults with overweight/obesity: a randomized controlled trial	Randomized, placebo-controlled, double-blind study	Inulin-type fructans with 6 g oligofructose + 2 g inulin from chicory root) 2 doses per day	Placebo not described
Dewulf et al. 2013	Insight into the prebiotic concept: lessons from an exploratory, double-blind intervention study with inulin-type fructans in obese women	A double blind, placebo controlled, intervention study	Inulin/oligofructose 50/50 mix (Synergy 1, namely, inulin/oligofructose 50/50 mix) 16 g/day	Maltodextrin
Valcheva et al. 2018	Inulin-type fructans improve active ulcerative colitis associated with microbiota changes and increased short-chain fatty acids levels	Randomized pilot study	Inulin type β-fructans (Synergy1, Orafti®) 7.5 or 15 g/day Orafti® Synergy1—50:50 inulin to	Placebo not detailed

Table 1 (continued)

			fructooligosaccharide mix; Beneo GmbH	
Costabile et al. 2010	A double-blind, placebo-controlled, cross-over study to establish the bifidogenic effect of a very-long-chain inulin extracted from globe artichoke (Cynara scolymus) in healthy human subjects	A double-blind, placebo-controlled, cross-over study Separated by a 3-week washout period	Inulin 10 g/day Extracted from globe artichoke root	Maltodextrin
Joossens et al. 2012	Effect of oligofructose-enriched inulin (OF-IN) on bacterial composition and disease activity of patients with Crohn's disease: results from a double-blinded randomized controlled trial	Single-center randomized placebo-controlled trial	Oligofructose-enriched inulin 10 g twice daily (Beneo GmbH)	Placebo 10 g twice daily

	Duration of exposure	Participants (*N*)	Microbiome assessment method	Microorganism identification reference
Vandeputte et al. 2017	4 weeks	42 healthy subjects	16S sequencing, V4 region	RDP classifier
Holscher et al. 2015	3 times of 21 days	29 healthy subjects	16S sequencing, V4 region	QIIME 1.8.0 Greengenes (version 13_8)
Baxter et al. 2019	2 weeks	49 healthy subjects	16S sequencing, V4 region	RDP classifier
Healey et al. 2018	3 weeks	34 healthy subjects with low dietary fiber (LDF) or high dietary fiber (HDF) intakes	16S sequencing, V3 and V4 regions	QIIME Greengenes (version 13_5)
Reimer et al. 2017	12 weeks	26 patients with overweight/obesity	16S sequencing, V3 and V4 regions	QIIME 1.9.1 Greengenes
Dewulf et al. 2013	3 months	30 obese women	Phylogenetic microarray	NA
Valcheva et al. 2018	9 weeks	25 patients with active ulcerative colitis	16S sequencing	QIIME Greengenes
Costabile et al. 2010	Two 3-week study periods	Thirty-two healthy adults	Fluorescence in situ hybridization	NA
Joossens et al. 2012	4 weeks	45 patients with inactive and mild to moderately active CD	16S sequencing	Not reported

Table 2 Key results: the associations of inulin supplementation with gut microbial diversity and abundance of specific microorganisms

	Total gut microbial diversity	Microorganism			
		Phylum	Class	Order	Family
Vandeputte et al. 2017	Increased Bray Curtis dissimilarity with inulin consumption	Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Firmicutes	Clostridia	Clostridiales	Lachnospiraceae
	Decreased observed richness	Proteobacteria	Proteobacteria	Desulfovibrionales	Desulfovibrionaceae
	No change in estimated richness (Chao1) and evenness, diversity (Shannon)				
Holscher et al. 2015	No significant differences in alpha diversity measures (observed species and phylogenetic diversity)	Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Proteobacteria	Deltaproteobacteria	Desulfovibrionales	Desulfovibrionaceae
	No significant change in the overall structure of the community	Firmicutes	Clostridia	Clostridiales	Lachnospiraceae
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
Baxter et al. 2019	Inulin altered the overall structure of the community	Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Firmicutes	Clostridia	Clostridiales	Lachnospiraceae
	No change in alpha diversity (inverse Simpson index)				
Healey et al. 2018	Large inter-individual variability in whole community microbiota responses to the inulin-type fructan prebiotic	Actinobacteria			
		Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Firmicutes			
		Firmicutes	Clostridia	Clostridiales	Lachnospiraceae
	No change in alpha diversity	Firmicutes	Clostridia	Clostridiales	Lachnospiraceae
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
		Firmicutes	Bacilli	Lactobacillales	Lactobacillaceae
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
Reimer et al. 2017	Overall microbial community structure differed before and after intervention	Actinobacteria			
		Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
	Decrease in alpha diversity (species richness), diversity				
	Chao1 metric not significant				
Dewulf et al. 2013	NA	Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
		Bacteroidetes	Bacteroidia	Bacteroidales	Bacteroidaceae
		Bacteroidetes	Bacteroidia	Bacteroidales	Bacteroidaceae
		Actinobacteria	Actinobacteria	Propionibacteriales	Propionibacteriaceae
		Firmicutes	Bacilli	Lactobacillales	Lactobacillaceae
Valcheva et al. 2018	No significant impact on the overall bacterial community structure in stool or biopsies.	Firmicutes	Clostridia	Clostridiales	Ruminococcaceae (stool of patients treated with 7.5 g/day)
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
		Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
		Firmicutes	Bacilli	Lactobacillales	Lachnospiraceae (stool of patients treated with 15 g/day)

Table 2 (continued)

		Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae (stool of patients treated with 15 g/day)
		Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Bacteroidetes	Bacteroidia	Bacteroidales	Porphyromonadaceae (taxa in biopsies from patients treated with 15 g/day dose)
		Bacteroidetes	Bacteroidia	Bacteroidales	Bacteroidaceae (taxa in biopsies from patients treated with 15 g/day dose)
		Bacteroidetes	Bacteroidia	Bacteroidales	Bacteroidaceae
		Bacteroidetes	Bacteroidia	Bacteroidales	Porphyromonadaceae
Costabile et al. 2010	NA	Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae
		Firmicutes	Bacilli	Lactobacillales	Lactobacillaceae
		Actinobacteria	Coriobacteriia	Coriobacteriales	Coriobacteriaceae
		Bacteroidetes	Bacteroidia	Bacteroidales	Bacteroidaceae
Joossens et al. 2012	Not reported	Firmicutes	Clostridia	Clostridiales	Ruminococcaceae
		Actinobacteria	Actinobacteria	Bifidobacteriales	Bifidobacteriaceae

	Microorganism		Abundance change attributed to intervention (if reported) and direction of change	
	Genus	Species		
Vandeputte et al. 2017	*Bifidobacterium*		Not reported	Increased
	Anaerostipes spp.		Not reported	Increased
	Bilophila		Not reported	Decreased
Holscher et al. 2015	*Bifidobacterium*		3- and 4-fold with 5.0 and 7.5 g agave inulin/day, respectively	Increased
	Bifidobacterium	*B. adolescentis*, *B. breve*, *B. longum*, *B. pseudolongum*		Increased
	Desulfovibrio		40%	Decreased
	Lachnobacterium		Only with 7.5 g agave inulin/day	Decreased
	Ruminococcus		Only with 7.5 g agave inulin/day	Decreased
Baxter et al. 2019	*Bifidobacterium*	4 different species	Fold change: 2.6 (seq040), 2.9 (seq064), 3.5 (seq024), 5.1 (seq184)	Increased
	Anaerostipes	2 different species	Fold change 2.1 (seq026) 2.2 (seq096)	Increased
Healey et al. 2018			10.98 ± 7.87 vs 19.95 ± 10.20 (mean ± SD, before vs after intervention)	Increased
	Bifidobacterium		6.69 ± 6.37 vs 15.07 ± 8.54 (mean ± SD, before vs after intervention)	Increased
			72.41 ± 10.75 vs 65.71 ± 11.03 (mean ± SD, before vs after intervention)	Decreased
	Coprococcus		4.50 ± 2.54 vs 3.55 ± 1.65	Decreased
	Dorea		1.75 ± 0.96 vs 1.20 ± 0.66	Decreased

Table 2 (continued)

	Ruminococcus		2.11 ± 1.59 vs 1.15 ± 1.04	Decreased
	Oscillospira		1.08 ± 0.53 vs 0.78 ± 0.46	Increased
	Lactobacillus		0.59 ± 1.29 vs 2.97 ± 5.64 in the LDF group	Increased
	Faecalibacterium		0.42 ± 0.18 vs 0.7 ± 0.28 in the HDF	Increased
Reimer et al. 2017			5.7% (2–9.3%) vs 17.4% (8.8–26.1%) (relative abundance, before vs after intervention)	Increased
	Bifidobacterium		3.6% (0.5–6.7%) vs 15.3% (6.2–24.3%)	Increased
Dewulf et al. 2013	*Bifidobacterium*		0.17 ± 0.19 vs 1.85 ± 2.15 (mean ± SD, before vs after intervention)	Increased
	Faecalibacterium prausnitzii		13.82 ± 7.79 vs 18.65 ± 4.85	Increased
	Bacteroides	*Bacteroides intestinalis*	2.09 ± 0.84 vs 1.54 ± 0.68	Decreased
	Bacteroides	*Bacteroides vulgatus*	4.20 ± 2.52 vs 2.57 ± 1.42	Decreased
	Propionibacterium		0.03 ± 0.03 vs 0.01 ± 0.01	Decreased
	Lactobacillus	*Lactobacillus gasseri*	0.01 ± 0.01 vs 0.27 ± 0.78	Increased
Valcheva et al. 2018				Increased
	Faecalibacterium (stool of patients treated with 7.5 g/day)			Increased
	Unclassified Ruminococcaceae (stool of patients treated with 7.5 g/day)			Increased
			9.9 ± 2.77 vs 17.7 ± 3.48 (before vs after)	Increased
			0.45 ± 0.28 vs 2.65 ± 1.36	Increased
	Bifidobacterium (stool of patients treated with 15 g/day)		0.46 ± 0.29 vs 2.44 ± 1.27	Increased
			1.99 ± 0.71 vs 0.46 ± 0.23	Decreased
			21.3 ± 2.37 vs 8.30 ± 2.70	Decreased
	Bacteroides (taxa in biopsies from patients treated with 15 g/day dose)		21.3 ± 2.37 vs 8.30 ± 2.70	Decreased
	Parabacteroides (taxa in biopsies from patients treated with 15 g/day dose)		1.51 ± 0.66 vs 0.29 ± 0.16	Decreased
Costabile et al. 2010	*Bifidobacterium*		2.82-fold	Increased
	Lactobacilli–enterococci		2.42-fold	Increased
	Atopobium		2.83-fold	Increased
	Bacteroides–Prevotella		1.77-fold	Decreased
Joossens et al. 2012	*Ruminococcus*	*Ruminococcus gnavus*	8.9 (7.9–9.3) vs 8.1 (7.6–9.2)	Decreased
	Bifidobacterium	*Bifidobacterium longum*	8.3 (6.6–8.9) vs 8.7 (7.5–9.3) (median number, before and after)	Increased

post prebiotic intervention samples clustering together, while others having their pre and post prebiotic intervention samples very distant from each other [18].

We observed several concordant changes between studies in the relative abundance of individual gut microbes after inulin exposure. The most consistent effect shown in all studies was an increase in the relative abundance of genus *Bifidobacterium* [15–20, 22, 23]. At species level, two studies reported increase in relative abundance of *Bifidobacterium longum* [16, 23], one study reported also increase in *Bifidobacterium adolescentis*, *Bifidobacterium breve*, and *Bifidobacterium pseudolongum* [16]. Importantly, one study reported an increase in *Bifidobacterium* in the stools of participants receiving 15 g/l of inulin [21]. Additionally, two studies showed an increase in *Anaerostipes* genus after inulin exposure [15, 17], three studies showed an increase in *Faecalibacterium* after inulin exposure [18, 20, 21], three studies showed an increase in *Lactobacillus* after inulin exposure [18, 20, 22], and three studies showed a decrease in *Bacteroides* after inulin exposure [20–22]. To note, the relative abundance of *Ruminococcus* increased in two studies after inulin exposure [16, 18], however, one study reported a decrease in *Ruminococcus gnavus*. Other changes in taxonomy were not consistent between the different published reports.

Discussion

Our systematic review assessed the evidence for the effects of inulin supplementation on the human gut microbiome. Overall, nine studies were available, and some concordant findings emerged between the published reports. In several studies, inulin supplementation disrupted the overall gut bacterial diversity, and altered the relative abundance of some taxa. The most consistent change was an increase in *Bifidobacterium*. This genus has the capacity to degrade oligofructose [17, 24]. Other concordant results included increase in relative abundance of *Anaerostipes*, *Faecalibacterium*, and *Lactobacillus*, and a decrease in relative abundance of *Bacteroides* after inulin supplementation. *Lactobacillus* and *Faecalibacterium* have demonstrated the ability to degrade oligofructose *in vitro* [25, 26]. Authors previously suggested that *Anaerostipes* may use lactate and acetate, produced by the increase in *Bifidobacterium* [27, 28], or may be able to metabolize inulin itself [17]. Moreover, it was previously reported that *Faecalibacterium prausnitzii* has the capacity to ferment inulin to butyrate [29]. Furthermore, the decrease in relative abundance of *Bacteroides* may be linked to a reduction in pH as a result of SCFA production [30].

Inulin is presented as a readily fermentable fiber by intestinal bacteria, generating large quantities of short-chain fatty acids (SCFA) [3]. However, this assertion was not confirmed in the *in vivo* studies reported here. Indeed, taxonomic changes induced by inulin supplementation were not associated with significant increases in fecal butyrate levels [17, 22]. Yet, these studies in humans found increase in members of Bifidobacteriaceae, Lachnospiraceae, and Ruminococcaceae families, which include most of known butyrate producers [17, 24, 29]. Thus, the link between stimulating butyrate production with inulin, supported by in vitro experiments, seems more complex and restricted in the *in vivo* scenario [17].

Our review and the different studies included in our review had several limitations that need to be highlighted. First, most of the studies used convenience sampling, with no sample size calculations reported, and several trials in human had small sample sizes. Second, several studies performed multiple statistical tests for taxonomic changes assessment, but did not perform correction for multiple testing. This may have overestimated the difference between the groups. Third, adverse events were inconsistently reported, whereas it is a major point in the safety evaluation of a product. Fourth, our search strategy was limited to peer-reviewed articles in the English language. Fifth, the methodology was highly variable among studies, and calculated pooled estimates of results were not feasible.

Importantly, the studies applied 16S sequencing that were unable to provide information at species and strains level, and were unable to provide microbial functional capacity. However, this information is important to evaluate the effects of a dietary intervention on the gut microbiome. Thus, future research should include metatranscriptomics, metaproteomics, and metabolomics to gain into in-depth effect of inulin supplementation. Another important finding of our review is the high inter-individual variation in response to the inulin exposure. Predict how the gut microbiome will respond to a dietary intervention is challenging, and controlling for potential cofounders, especially diet and medications, in large trial will be pivotal to accurately assess effect of inulin on the human gut microbiome and its implication in health [31, 32].

Conclusions

Our systematic review assessed the evidence for the effects of inulin supplementation on the human gut microbiome. The most consistent change was an increase in *Bifidobacterium*. Other concordant results included an increase in relative abundance of *Anaerostipes*, *Faecalibacterium*, and *Lactobacillus*, and a decrease in relative abundance of *Bacteroides* after inulin supplementation. However, these *in vivo* studies did not confirm *in vitro* experiments as the taxonomic alterations were not associated with increase in SCFA levels.

Authors' contributions Q.L.B, G.C., F.J, D.L., E.B., and E.M. directly participated in study design and protocol preparation. Q.L.B and E.M. screened abstracts and titles for inclusion. Q.L.B, G.C., F.J, and E.M.

participated in review of full-text articles. E.M. drafted the manuscript. All authors participated in manuscript editing and critical review.

Data availability Data from the study are available upon request from the corresponding author (E Montassier).

Compliance with ethical standards

Competing interests The authors declare that they have no competing interests.

Ethics approval and consent to participate Not applicable.

Abbreviations OTU, Operational taxonomic unit; SCFA, Short-chain fatty acids

References

1. Gibson GR, Roberfroid MB (1995) Dietary modulation of the human colonic microbiota: introducing the concept of prebiotics. J Nutr 125:1401–1412
2. Ellegard L, Andersson H, Bosaeus I (1997) Inulin and oligofructose do not influence the absorption of cholesterol, or the excretion of cholesterol, Ca, Mg, Zn, Fe, or bile acids but increases energy excretion in ileostomy subjects. Eur J Clin Nutr 51:1–5
3. Roberfroid M, Slavin J (2000) Nondigestible oligosaccharides. Crit Rev Food Sci Nutr 40:461–480
4. Micka A, Siepelmeyer A, Holz A, Theis S, Schön C (2017) Effect of consumption of chicory inulin on bowel function in healthy subjects with constipation: a randomized, double-blind, placebo-controlled trial. Int J Food Sci Nutr 68:82–89
5. EFSA Panel on Dietic Products Nutrition and Allergies (NDA) (2015) Scientific opinion on the substantiation of a health claim related to “native chicory inulin” and maintenance of normal defecation by increasing stool frequency pursuant to Article of Regulation (EC) No 1924/2006. EFSA J 13:3951
6. Ford AC, Moayyedi P, Chey WD, Harris LA, Lacy BE, Saito YA, Quigley EMM, ACG task force on management of irritable bowel syndrome (2018) American College of Gastroenterology Monograph on Management of Irritable Bowel Syndrome. Am J Gastroenterol 113:1–18
7. Singh V, Yeoh BS, Walker RE (2019) Microbiota fermentation-NLRP3 axis shapes the impact of dietary fibres on intestinal inflammation. Gut. 316250
8. Bafeta A, Koh M, Riveros C, Ravaud P (2018) Harms reporting in randomized controlled trials of interventions aimed at modifying microbiota: a systematic review. Ann Intern Med 169:240–247
9. Durack J, Lynch SV (2019) The gut microbiome: relationship with disease and opportunities for therapy. J Exp Med 216:20–40
10. Shmagel A, Demmer R, Knights D, Butler M, Langsetmo L, Lane NE, Ensrud K (2019) Effects of glucosamine and chondroitin sulfate on gut microbial composition: a systematic review of evidence from animal and human studies. Nutrients. 11:2
11. Le Bastard Q, Al-Ghalith GA, Grégoire M, Chapelet G, Javaudin F, Dailly E, Batard E, Knights D, Montassier E (2018) Systematic review: human gut dysbiosis induced by non-antibiotic prescription medications. Aliment Pharmacol Ther 47:332–345
12. Claesson MJ, Clooney AG, O’Toole PW (2017) A clinician’s guide to microbiome analysis. Nat Rev Gastroenterol Hepatol 14:585–595
13. Tyler AD, Smith MI, Silverberg MS (2014) Analyzing the human microbiome: a "how to" guide for physicians. Am J Gastroenterol 109:983–993
14. https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi. Accessed May 2019.
15. Vandeputte D, Falony G, Vieira-Silva S, Wang J, Sailer M, Theis S, Verbeke K, Raes J (2017) Prebiotic inulin-type fructans induce specific changes in the human gut microbiota. Gut. 66:1968–1974
16. Holscher HD, Bauer LL, Gourineni V, Pelkman CL, Fahey GC Jr, Swanson KS (2015) Agave inulin supplementation affects the fecal microbiota of healthy adults participating in a randomized, double-blind, placebo-controlled, crossover trial. J Nutr 145:2025–2032
17. Baxter NT, Schmidt AW, Venkataraman A, Kim KS, Waldron C, Schmidt TM (2019) Dynamics of human gut microbiota and short-chain fatty acids in response to dietary interventions with three fermentable fibers. MBio. 10:e02566–e02518
18. Healey G, Murphy R, Butts C, Brough L, Whelan K, Coad J (2018) Habitual dietary fibre intake influences gut microbiota response to an inulin-type fructan prebiotic: a randomised, double-blind, placebo-controlled, cross-over, human intervention study. Br J Nutr 119: 176–189
19. Reimer RA, Willis HJ, Tunnicliffe JM, Park H, Madsen KL, Soto-Vaca A (2017) Inulin type fructans and whey protein both modulate appetite but only fructans alter gut microbiota in adults with overweight/obesity: A randomized controlled trial. Mol Nutr Food Res 61
20. Dewulf EM, Cani PD, Claus SP, Fuentes S, Puylaert PG, Neyrinck AM, Bindels LB, de Vos WM, Gibson GR, Thissen JP, Delzenne NM (2013) Insight into the prebiotic concept: lessons from an exploratory, double blind intervention study with inulin-type fructans in obese women. Gut. 62:1112–1121
21. Valcheva R, Koleva P, Martínez I, Walter J, Gänzle MG, Dieleman LA (2019) Inulin-type fructans improve active ulcerative colitis associated with microbiota changes and increased short-chain fatty acids levels. Gut Microbes 10:334–357
22. Costabile A, Kolida S, Klinder A, Gietl E, Bäuerlein M, Frohberg C, Landschütze V, Gibson GR (2010) A double-blind, placebo-controlled, cross-over study to establish the bifidogenic effect of a very-long-chain inulin extracted from globe artichoke (Cynara scolymus) in healthy human subjects. Br J Nutr 104:1007–1017
23. Joossens M, De Preter V, Ballet V, Verbeke K, Rutgeerts P, Vermeire S (2012) Effect of oligofructose-enriched inulin (OFIN) on bacterial composition and disease activity of patients with Crohn’s disease: results from a double-blinded randomised controlled trial. Gut. 61:958
24. Rossi M, Corradini C, Amaretti A, Nicolini M, Pompei A, Zanoni S, Matteuzzi D (2005) Fermentation of fructooligosaccharides and inulin by bifidobacteria: a comparative study of pure and fecal cultures. Appl Environ Microbiol 71:6150–6158
25. Hughes SA, Shewry PR, Li L, Gibson GR, Sanz ML, Rastall RA (2007) In vitro fermentation by human faecal microflora of wheat arabinoxylans. J Agric Food Chem 55:4589–4595
26. Kolida S, Gibson GR (2007) Prebiotic capacity of inulinfructans. J Nutr 137:2503–2506
27. Allen-Vercoe E, Daigneault M, White A, Panaccione R, Duncan SH, Flint HJ, O’Neal L, Lawson PA (2012) Anaerostipes hadrus comb. nov., a dominant species within the human colonic microbiota; reclassification of Eubacterium hadrum Moore et al. 1976. Anaerobe. 18:523–529
28. Duncan SH, Louis P, Flint HJ (2004) Lactate-utilizing bacteria, isolated from human feces, that produce butyrate as a major fermentation product. Appl Environ Microbiol 70:5810–5817
29. Moens F, De Vuyst L (2007) Inulin-type fructan degradation capacity of Clostridium cluster IV and XIVa butyrate-producing colon bacteria and their associated metabolic outcomes. Benefic Microbes 8:473–490

30. Deehan EC, Duar RM, Armet AM, Perez-Muñoz ME, Jin M, Walter J (2017) Modulation of the gastrointestinal microbiome with nondigestible fermentable carbohydrates to improve human health. Microbiol Spectr 5:5
31. Zhernakova A, Kurilshikov A, Bonder MJ, Tigchelaar EF, Schirmer M, Vatanen T, Mujagic Z, Vila AV, Falony G, Vieira-Silva S et al (2016) Population-based metagenomics analysis reveals markers for gut microbiome composition and diversity. Science. 352:565–569
32. Falony G, Joossens M, Vieira-Silva S, Wang J, Darzi Y, Faust K, Kurilshikov A, Bonder MJ, Valles-Colomer M, Vandeputte D (2016) Population-level analysis of gut microbiome variation. Science. 352:560–564

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